

Maclaurin Series & Convergence of Series

ANSWER KEY



Constructing a Maclaurin series

- Find the first four non-zero terms of the Maclaurin series for $f(x) = e^x$, using $f(x) = f(0) + f'(0)x + (f''(0)/2!)x^2 + (f'''(0)/3!)x^3 + \dots$.
Since every derivative of e^x equals e^x , and $e^0 = 1$: all derivatives at 0 equal 1. So $e^x \approx 1 + x + x^2/2! + x^3/3! = 1 + x + x^2/2 + x^3/6$.
- Find the Maclaurin series for $f(x) = \sin x$ up to the term in x^5 .
 $f(0) = 0$, $f'(0) = \cos 0 = 1$, $f''(0) = -\sin 0 = 0$, $f'''(0) = -\cos 0 = -1$, $f^{(4)}(0) = \sin 0 = 0$, $f^{(5)}(0) = \cos 0 = 1$. Series: $\sin x \approx x - x^3/3! + x^5/5! = x - x^3/6 + x^5/120$.

Using known series to build new ones

- Using the Maclaurin series for e^x , find the Maclaurin series for e^{-2x} up to the term in x^3 .
Substitute $-2x$ for x in $e^x \approx 1 + x + x^2/2 + x^3/6$:
 $e^{-2x} \approx 1 + (-2x) + (-2x)^2/2 + (-2x)^3/6 = 1 - 2x + 2x^2 - 4x^3/3$.
- Given the Maclaurin series $\sin x \approx x - x^3/6$ and $\cos x \approx 1 - x^2/2$, find the Maclaurin series for $\tan x$ up to the term in x^3 , using $\tan x = \sin x / (\cos x)$ and polynomial division.
 $(x - x^3/6) / (1 - x^2/2) \approx (x - x^3/6)(1 + x^2/2)$ (using $1/(1-u) \approx 1+u$ for small u)
 $= x + x^3/2 - x^3/6 + O(x^5) = x + x^3/3$.

Using a Maclaurin series to approximate a value

- Use the Maclaurin series for e^x up to the x^3 term to approximate $e^{0.1}$, to four decimal places, and compare with the true value $e^{0.1} \approx 1.10517$.
 $e^{0.1} \approx 1 + 0.1 + (0.1)^2/2 + (0.1)^3/6 = 1 + 0.1 + 0.005 + 0.0001\bar{6} \approx 1.1052$. This is very close to the true value 1.10517, with an error of about 0.00003.

Tests for convergence of a series

- Use the ratio test to determine whether the series $1/2 + 2/4 + 3/8 + 4/16 + \dots + n/2^n + \dots$ converges.
Ratio test: $\lim_{n \rightarrow \infty} |a_{n+1}/a_n| = \lim_{n \rightarrow \infty} (n+1)/2^{n+1} / (n/2^n) = \lim_{n \rightarrow \infty} (n+1)/(2n) = 1/2$. Since $1/2 < 1$, the series converges.

- 7 Use the ratio test to determine whether the Maclaurin series for e^x , $1 + x + x^2/2! + x^3/3! + \dots + x^n/n! + \dots$, converges for all real x .
 Ratio test: $\lim_{n \rightarrow \infty} |x^{n+1}/(n+1)! / (x^n/n!)| = \lim_{n \rightarrow \infty} |x|/(n+1) = 0$ for any fixed x . Since the limit is $0 < 1$ regardless of x , the series converges for all real values of x .

Determining the radius of convergence

- 8 Use the ratio test to find the radius of convergence of the power series $x + x^2/2 + x^3/3 + \dots + x^n/n + \dots$.
 Ratio test: $\lim_{n \rightarrow \infty} |x^{n+1}/(n+1)/(x^n/n)| = \lim_{n \rightarrow \infty} n|x|/(n+1) = |x|$. The series converges when $|x| < 1$, so the radius of convergence is $R = 1$.
- 9 Find the radius of convergence of the series $1 + x + 2x^2 + 6x^3 + \dots + n! x^n + \dots$, and explain what this tells us about where the series is useful.
 Ratio test: $\lim_{n \rightarrow \infty} |(n+1)! x^{n+1}/(n! x^n)| = \lim_{n \rightarrow \infty} (n+1)|x| = \infty$ for any $x \neq 0$. Since this limit is never less than 1 (except at $x = 0$), the radius of convergence is $R = 0$ — the series only converges at the single point $x = 0$, making it useless as an approximation anywhere else.

Comparing a Maclaurin approximation to the exact function

- 10 Using the Maclaurin series $\cos x \approx 1 - x^2/2 + x^4/24$, approximate $\cos(0.3)$ to four decimal places, and compare with the true value $\cos(0.3) \approx 0.9553$.
 $\cos(0.3) \approx 1 - (0.3)^2/2 + (0.3)^4/24 = 1 - 0.045 + 0.0003375 \approx 0.9553$. This matches the true value to four decimal places, showing the Maclaurin approximation is highly accurate for small x .

Maclaurin series for $\ln(1+x)$

- 11 Find the Maclaurin series for $f(x) = \ln(1+x)$ up to the term in x^4 , using $f(x) = f(0) + f'(0)x + (f''(0)/2!)x^2 + (f'''(0)/3!)x^3 + (f^{(4)}(0)/4!)x^4$.
 $f(0) = \ln 1 = 0$. $f'(x) = (1+x)^{-1}$, so $f'(0) = 1$. $f''(x) = -(1+x)^{-2}$, so $f''(0) = -1$. $f'''(x) = 2(1+x)^{-3}$, so $f'''(0) = 2$. $f^{(4)}(x) = -6(1+x)^{-4}$, so $f^{(4)}(0) = -6$. Series:
 $\ln(1+x) \approx x - x^2/2 + 2x^3/6 - 6x^4/24 = x - x^2/2 + x^3/3 - x^4/4$.

Determining the interval of convergence using the ratio test

- 12 Find the radius of convergence of the Maclaurin series for $\ln(1+x)$, $x - x^2/2 + x^3/3 - \dots + (-1)^{n+1}x^n/n + \dots$, using the ratio test.
 Ratio test: $\lim_{n \rightarrow \infty} |x^{n+1}/(n+1)/(x^n/n)| = \lim_{n \rightarrow \infty} n|x|/(n+1) = |x|$. The series converges when $|x| < 1$, so the radius of convergence is $R = 1$ (consistent with the earlier result found for a structurally similar series).

Combining Maclaurin series through multiplication

- 13** Using the Maclaurin series $e^x \approx 1 + x + x^2/2$ and $1/(1-x) \approx 1 + x + x^2$ (valid for small x), find the Maclaurin series for $e^x/(1-x)$ up to the term in x^2 , by multiplying the two series together.

Multiply term by term, keeping only terms up to x^2 :

$$(1 + x + x^2/2)(1 + x + x^2) = 1 + x + x^2 + x + x^2 + x^2/2 + O(x^3) = 1 + 2x + 5x^2/2 + O(x^3). \text{ So } e^x/(1-x) \approx 1 + 2x + 5/2x^2.$$

Using differentiation to find a Maclaurin series

- 14** Given the Maclaurin series $1/(1-x) \approx 1 + x + x^2 + x^3 + \dots$ for $|x| < 1$, differentiate both sides term by term to find the Maclaurin series for $1/(1-x)^2$ up to the term in x^2 .

Differentiating $1/(1-x)$ gives $1/(1-x)^2$ on the left. Differentiating the right side term by term:

$$d/(dx)(1 + x + x^2 + x^3 + \dots) = 1 + 2x + 3x^2 + \dots. \text{ So } 1/(1-x)^2 \approx 1 + 2x + 3x^2.$$

Estimating an error bound in a Maclaurin approximation

- 15** The Maclaurin series for $\sin x$ truncated after the x^3 term is $\sin x \approx x - x^3/6$. Use this to approximate $\sin(0.2)$, and find the absolute error compared to the true value $\sin(0.2) = 0.1986693\dots$
 $\sin(0.2) \approx 0.2 - (0.2)^3/6 = 0.2 - 0.008/6 = 0.2 - 0.0013333 = 0.1986667$. Absolute error:
 $|0.1986693 - 0.1986667| \approx 0.0000027$, showing that even this low-order Maclaurin approximation is extremely accurate for small values of x close to 0 — the error here is on the order of 10^{-6} .
- 16** Find the first three non-zero terms of the Maclaurin series for $f(x) = (1+x)^{1/2}$, using the binomial series $(1+x)^n \approx 1 + nx + n(n-1)/2x^2$ with $n = 1/2$.
Substituting $n = 1/2$: $(1+x)^{1/2} \approx 1 + (1/2)x + ((1/2)(-1/2))/2x^2 = 1 + (1/2)x - (1/8)x^2$.
- 17** Use the Maclaurin series for e^x up to the x^4 term to approximate $\lim_{x \rightarrow 0} (e^x - 1 - x)/x^2$, and check that your answer matches the value obtained using L'Hopital's rule.
Using $e^x \approx 1 + x + x^2/2 + x^3/6 + \dots$: $e^x - 1 - x \approx x^2/2 + x^3/6 + \dots$. Dividing by x^2 :
 $(e^x - 1 - x)/x^2 \approx 1/2 + x/6 + \dots$, which tends to $1/2$ as $x \rightarrow 0$. Using L'Hopital's rule twice on the original limit (since it is initially $0/0$) gives $\lim_{x \rightarrow 0} (e^x - 1)/(2x) = e^0/2 = 1/2$, confirming the same result.
- 18** This question has three parts. (a) Find the Maclaurin series for $f(x) = \cos(x^2)$ up to the term in x^8 , by substituting x^2 into the known series $\cos u \approx 1 - u^2/2 + u^4/24$. (b) Use your result to write down an approximation for $\int_0^1 \cos(x^2) dx$ as a sum of simple fractions (this integral has no elementary closed form, so series approximation is genuinely useful here). (c) Evaluate this approximation as a single decimal, to three decimal places.
(a) Substituting $u = x^2$: $\cos(x^2) \approx 1 - (x^2)^2/2 + (x^2)^4/24 = 1 - x^4/2 + x^8/24$. (b) Integrating term by term: $\int (1 - x^4/2 + x^8/24) dx = x - x^5/10 + x^9/216 + C$. Evaluating from $x = 0$ to $x = 1$ gives $1 - 1/10 + 1/216$. (c) $1 - 0.1 + 0.00463 \approx 0.905$.

Checking endpoints of an interval of convergence

- 19** The Maclaurin series for $\ln(1+x)$ has radius of convergence $R = 1$ (found earlier in this worksheet). Explain why checking the endpoints $x = 1$ and $x = -1$ separately is necessary to determine the full interval of convergence, given that at $x = 1$ the series becomes $1 - 1/2 + 1/3 - 1/4 + \dots$ (the alternating harmonic series, which is known to converge).

The ratio test only determines where a power series converges absolutely (the open interval $|x| < 1$); it says nothing about the boundary points $x = \pm 1$, since the ratio test's limit equals exactly 1 there (inconclusive), so each endpoint must be checked separately by direct substitution. At $x = 1$, the series becomes $1 - 1/2 + 1/3 - 1/4 + \dots$, the alternating harmonic series, which converges by the alternating series test (its terms decrease steadily to 0), so $x = 1$ is included. At $x = -1$, the series becomes $-1 - 1/2 - 1/3 - 1/4 - \dots$, the negative of the (divergent) harmonic series, so $x = -1$ is excluded. The full interval of convergence is therefore $-1 < x \leq 1$.

Using a Maclaurin series to evaluate a standard trigonometric limit

- 20** Use the Maclaurin series for $\sin x$ up to the x^3 term to evaluate $\lim_{x \rightarrow 0} (\sin x)/x$.
Using $\sin x \approx x - x^3/6$: $(\sin x)/x \approx (x - x^3/6)/x = 1 - x^2/6$. As $x \rightarrow 0$, this tends to 1, matching the well-known standard limit $\lim_{x \rightarrow 0} (\sin x)/x = 1$.
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Identifying the general term of a Maclaurin series

- 21** Which of the following is the general term of the Maclaurin series for $\cos x$?

a. $(-1)^n x^{2n}/(2n)!$ ✓

b. $(-1)^n x^n/n!$

c. $x^{2n}/(2n)!$

d. $(-1)^{n+1} x^{2n-1}/(2n-1)!$

The Maclaurin series for $\cos x$ is $1 - x^2/2! + x^4/4! - x^6/6! + \dots$, which has only even powers of x with alternating signs and factorial denominators matching the power. Checking $(-1)^n x^{2n}/(2n)!$ at $n = 0$ gives 1, at $n = 1$ gives $-x^2/2!$, and at $n = 2$ gives $x^4/4!$ — all correct. The second option is the general term for e^x with alternating signs, not cosine's even-power structure; the third omits the alternating signs; the fourth describes $\sin x$'s general term, not $\cos x$'s.

Comparing the accuracy of two Maclaurin approximations

- 22** Both $e^x \approx 1 + x + x^2/2$ and $\cos x \approx 1 - x^2/2$ are second-order Maclaurin approximations. Without calculating exact values, explain which approximation is likely more accurate at $x = 1$, referencing the size of the next omitted term in each series.

For e^x , the next omitted term is $x^3/6$, which at $x = 1$ equals $1/6 \approx 0.167$. For $\cos x$, the next omitted term is $x^4/24$, which at $x = 1$ equals $1/24 \approx 0.042$. Since the cosine approximation's next omitted term is smaller, the $\cos x$ approximation is likely more accurate at $x = 1$ than the e^x approximation.